CyberTraining Workshop: Completed Work and Preliminary Project Status

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1 Introductory Remarks

Modeling excited states and their associated dynamics firmly remains a focal area of research in computational chemistry. The breadth of research subjects spanned by quantum dynamics undeniably contributes to this interest. From photochemistry and rate theory to materials design and quantum information science (QIS), quantum dynamics simulations offer insight into significant areas of research across chemical disciplines. Moreover, advancements across general computational science—including machine learning methods and quantum computing architectures—pave the way for increased resources for larger, more accurate quantum dynamics simulations. Thus, understanding the theoretical foundations and the practical implementations of excited state methods and nonadiabatic dynamics is of substantial importance for aspiring computational chemists.

The 2026 CyberTraining Workshop Summer School on Modeling Excited States was an excellent experience allowing for the teaching of such methodologies. Through guest speakers and hands-on tutorials, participants were able to gain knowledge on fundamentals of nonadiabatic dynamics and quantum methods as well as explore state-of-the-art software packages in an HPC environment. Numerous calculations and computations were performed via ChemML[4], PySCF[13], PySpawn[3], Libra[12], and Tenso[11]. The remainder of this report will highlight selected calculations conducted beyond the workshop exercises. Furthermore, the utilization of these software tools for a project inspired by QIS studies will also be discussed. Preliminary goals and results will be provided, and a course for future computations will be given.

2 Summary of Completed Lessons and Outcomes

The directories enclosed in the repository include the major tutorials completed during the hands-on sessions of the workshop. While most Libra and Tenso tutorials were completed on a local device, the remainder of the calculations were carried out on CCR resources.

In the PySCF/Prism tutorial, this included the given tutorial examples as well as several single reference (i.e. HF/DFT) and multireference calculations for the investigation of nitroxyl. Moreover, further work was done to determine the effect of different DFT guesses on the final CASCI energies of a molecule. In the example of ethylene, it was seen that LDA, PBE, B3LYP, and a user-defined range-separated hybrid (RSH) functional all yielded results that—while substantially different from the HF-guess orbitals—are very similar to other DFT results. This is especially true for all functionals beyond LDA, suggesting that using a "cheap" functional (i.e. GGA) can potentially yield the same results as an "expensive" functional (i.e. hybrids, range-separated). Such a conclusion will require further testing as well as deployment to condensed-phase systems. Notably, solvation models—including PCM—were also implemented in composed input files and ran. Several jobs related to the project were also carried out and will be discussed in the succeeding section. OpenMolcas and PySpawn were also utilized to run several calculations beyond the tutorials. These include hessian conversion to PySpawn readable files; CASSCF and RASSCF calculations on excited states for doublet-ground state molecules; and completion of tutorial files with orbital adjustments. Moreover, both packages were used in the project. Furthermore, Tenso calculations were carried out to determine the effect of frequency, temperature, width, and Brownian motion on the spin boson models. Moreover, all completed regular tutorial files for each package (unless ran locally) are in the respective directory.

3 Project: Overview and Initial Findings

There is a growing research interest in designing electron spin molecular qubits for QIS applications including quantum sensing and quantum communication. In order to accomplish practical implementation of electron spin molecular qubits, molecules must be designed to have sufficient spin coherence times and appropriate electronic structures for useful QIS manipulation. Thus, having the ability to accurately model such excited state energy levels and dynamics of qubit candidates is of supreme interest.

The focus of this project is to utilize electronic structure packages and quantum dynamics to accurately predict and model excited state dynamics in potential qubit candidates. Such qubit candidates will be inspired by work such as Wasielewski et al.[7, 8, 10, 9, 6], where chromophore- and radical-based molecules are employed for QIS candidacy.

The first phase of this project is the analysis of a test molecule and the successful utilization of the required softwares. Since many examined qubit candidates feature aromatic cyclic groups, cyclopentadiene was chosen as a candidate. The molecule was optimized with ωb97xd/aug-cc-pVDZ on home resources, and the hessian was calculated with ORCA. Using CCR resources, PySCF excited state calculations with TDA/TDDFT were performed on cyclopentadiene using the B3LYP and a user-defined RSH functional ($\alpha = 0.25, \omega = 200$). The results were compared to QUEST data.[5] While B3LYP seemingly underestimated the

Transition	f	E(eV)
ppi	0.084	5.540
p3s	0.000	5.775
p3p	0.037	6.407
p3p	0.000	6.449
p3p	0.046	6.556
ppi	0.010	6.451
ppi	0.000	3.308
ppi	0.000	5.117
p3s	0.000	5.725
p3p	0.000	6.361

Table 1: QUEST Data for cyclopentadiene

	TDA-RSH		TDDFT-RSH		TDA-B3LYP		TDDFT-B3LYP	
Energy Level	f	E(eV)	f	E(eV)	f	E(eV)	f	E(eV)
Ex. State 1	0.132	5.48098	0.084	5.12390	0.000	5.13874	0.081	5.00154
Ex. State 2	0.000	5.66810	0.000	5.66730	0.124	5.35556	0.000	5.13806
Ex. State 3	0.038	6.31477	0.037	6.31334	0.033	5.70560	0.032	5.70430
Ex. State 4	0.000	6.35774	0.000	6.35693	0.000	5.72957	0.000	5.72897
Ex. State 5	0.061	6.54483	0.054	6.53939	0.066	5.98514	0.053	5.97844
Ex. State 6	0.003	6.70859	0.002	6.69052	0.000	6.29780	0.000	6.29448

Table 2: TDA and TDDFT Energies for cyclopentadiene with B3LYP and RSH Functionals

energies of excited states—especially higher energy ones—RSH performed quite well, often matching energies within 0.1 eV. Notably, RSH-TDDFT performed slightly better in calculating oscillator strengths than RSH-TDA.

Furthermore, the hessian of cyclopentadiene was converted into the .hdf5 necessary for PySpawn simulations. OpenMolcas was utilized to calculate the input orbitals (INPORB) for the PySpawn calculation. The excited state calculation was performed for the first singlet excited state. PySpawn simulation was then performed, and the S1 population decay was captured as presented in Figure 1.

Overall, the analysis of cyclopentadiene is a useful test for establishing a workflow for combining TDDFT methods with PySpawn. Future work will look at integrating PySpawn into ORCA / Q-Chem software workflows for DFT-centered PySpawn runs. By using high-level DFT methods like SRSH-PCM[1, 2], accurate excited states can be captured while lowering computational costs from CAS-based techniques. Moreover, this will allow for the investigation of larger, more complex molecules.

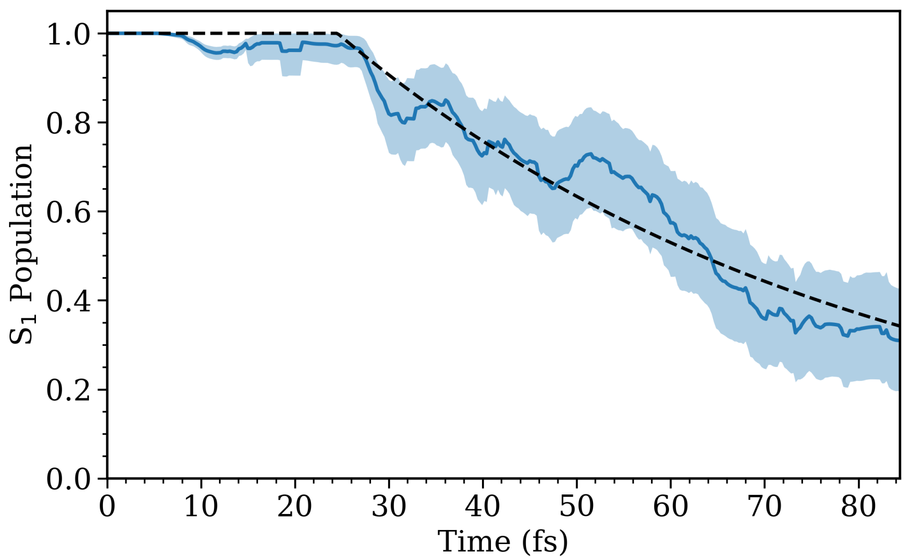

Figure 1: S1 Population Decay for Cyclopentadiene.

4 Concluding Remarks and Future Aspirations

Quantum dynamics methods are a central field of computational chemistry and an indispensable manner of studying quantum phenomena. Through the CyberTraining Workshop, valuable training on excited-state and nonadiabatic dynamics was transmitted to the audience through theory lectures and software demos. Such sessions allowed for the completion of numerous simulations on software packages including Libra, Tenso, PySCF, PySpawn, and more. Furthermore, the knowledge gained from the workshop was used to design a project using the examined software and conduct preliminary investigations.

5 References

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